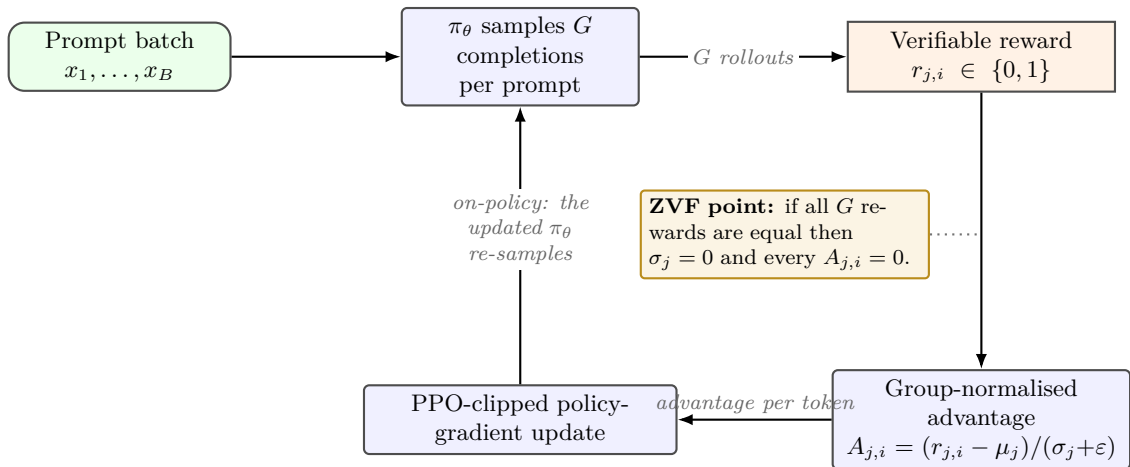
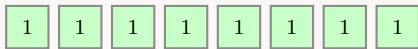




The ZVF point, in detail

Degenerate group: $\sigma_j = 0$

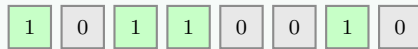


$$\mu_j = 1.00 \quad \sigma_j = 0.00$$

$$\Rightarrow A_{j,i} = (1 - 1) / (0 + \epsilon) = 0 \text{ for every } i$$

The surrogate term vanishes, so the group contributes *no* reward-relative gradient. The KL pull toward π_{ref} still applies, so the step spends rollout compute as a pure pull back to the reference. The symmetric case (all rewards 0) behaves identically.

Informative group: $\sigma_j > 0$



$$\mu_j = 0.50 \quad \sigma_j \approx 0.53$$

$$\Rightarrow A_{j,i} \approx \pm 0.94: \text{ both outcomes receive gradient}$$

Correct and incorrect completions both receive a reward-relative gradient, signed by whether the completion beat its own group. This is the signal the recipe exists to produce; a batch's ZVF is the fraction of groups that never reach it.

